

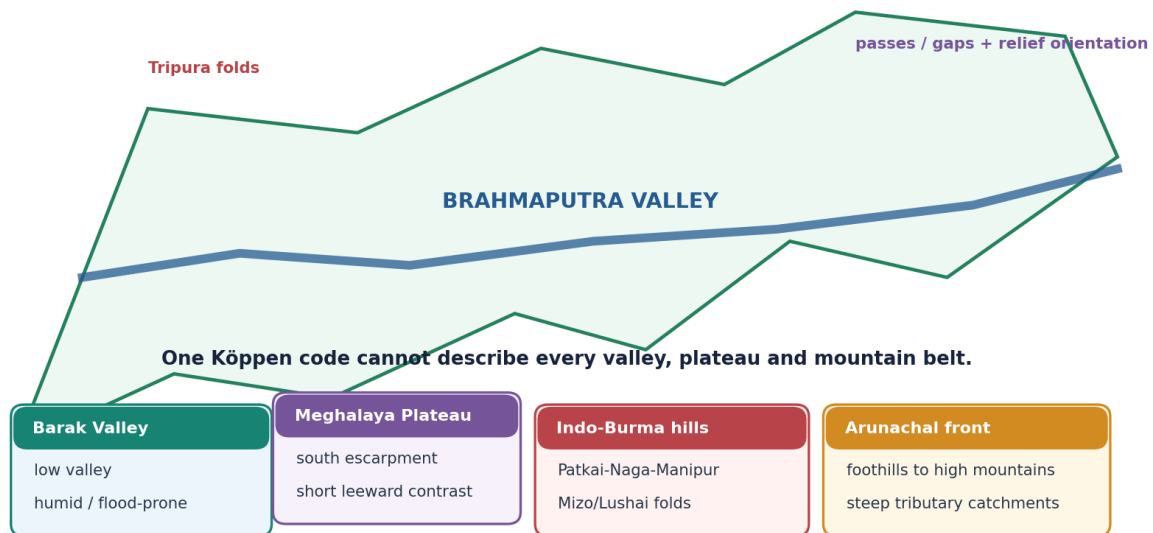
SOLVED PRACTICE WORKBOOK

Geography 21 - India Humid Subtropical Northeast - Solved Practice Workbook

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing

Northeast India: humid-subtropical mosaic and physical frame

Original schematic; not to scale



Original deterministic study visual; labels are explanatory, not textual quotations.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

Part II - Premium solved practice workbook

Workbook rules

- Correct choices rotate continuously **A -> B -> C -> D**.
- "Officially verified" is used only where a local official key was available.
- Missing objective keys are retained as **INFERRED ANSWER - NOT OFFICIALLY VERIFIED**, with confidence and elimination.
- Every Mains solution follows **claim -> named evidence/example -> analysis -> qualification -> verdict** and ends with **Why this earns marks**.

A. Solved Prelims PYQs and advanced MCQs

Q1. UPSC Prelims 2024, GS-I, Set A, Q7 - officially verified

Consider the following statements:

Statement-I: Rainfall is one of the reasons for weathering of rocks. **Statement-II:** Rain water contains carbon dioxide in solution. **Statement-III:** Rain water contains atmospheric oxygen.

Which one of the following is correct in respect of the above statements?

- A. Both Statement-II and Statement-III are correct and both of them explain Statement-I
B. Both Statement-II and Statement-III are correct, but only one of them explains Statement-I
C. Only one of Statements II and III is correct and that explains Statement-I
D. Neither Statement-II nor Statement-III is correct

Answer: A - officially verified from the locally available official Set-A key.

Explanation: Rain provides water for physical and chemical weathering. Dissolved carbon dioxide forms weak carbonic acid, aiding carbonate and silicate reactions. Dissolved oxygen supports oxidation. In the humid Northeast, high water availability accelerates weathering, but slope erosion may remove profiles before deep soil develops.

Q2. Original - classification

Which is the most defensible statement about Köppen Cwa/Cfa labels in Northeast India?

- A. Cwa applies to every valley because all winter months are rainless.
B. The labels require station thresholds, while highlands and tropical valleys may fall outside either label.
C. Cfa is determined solely by annual rainfall above 200 cm.
D. Any tea-growing district is automatically Cwa.

Answer: B.

Explanation: Köppen classes use monthly temperature and precipitation rules. Annual rainfall, crop type or state location alone cannot assign a code.

Q3. Original - orographic rainfall

With reference to the very wet southern slopes of Meghalaya, which sequence is most appropriate?

- A. Westerly dry air -> downslope warming -> frontal snowfall
B. Bay moisture -> plateau descent -> evaporation -> rain maximum
C. Moist low-level flow -> abrupt uplift -> cooling/condensation -> terrain-convection enhancement
D. High altitude alone -> rain independent of wind direction

Answer: C.

Explanation: Moisture supply, favourable flow orientation, sharp uplift and convective/cloud processes combine. Altitude without moist inflow is insufficient.

Q4. UPSC Prelims 2023, GS-I, Q63 - key unavailable locally

Consider the following statements:

Statement-I: The soil in tropical rain forests is rich in nutrients. **Statement-II:** The high temperature and moisture of tropical rain forests cause dead organic matter in the soil to decompose quickly.

Which one of the following is correct in respect of the above statements?

A. Both Statement-I and Statement-II are correct and Statement-II is the correct explanation for Statement-I B. Both Statement-I and Statement-II are correct and Statement-II is not the correct explanation for Statement-I C. Statement-I is correct but Statement-II is incorrect D. Statement-I is incorrect but Statement-II is correct

INFERRED ANSWER - NOT OFFICIALLY VERIFIED: D. Confidence: high, subject to the exact official-set wording/key.

Elimination: Nutrients are rapidly recycled and concentrated in biomass rather than forming a universally nutrient-rich soil. Warmth and moisture accelerate decomposition. Statement-I is therefore incorrect and Statement-II correct, routing to D. Retain the inferred label because no official local key was available.

Q5. Original - flood processes

Which intervention best addresses **bank erosion**, rather than only temporary inundation?

A. Repeated bankline mapping combined with setbacks, selective protection and rights-based resettlement B. Increasing the number of rain gauges only C. Pumping water from all wetlands D. Declaring every floodplain a permanent no-use zone without tenure safeguards

Answer: A.

Explanation: Erosion removes land and shifts the channel edge. Forecasting helps, but land-loss policy, bankline monitoring and selective engineering are central.

Q6. Original - Brahmaputra causation

Which combination best explains recurrent Brahmaputra flood and erosion risk?

A. Deforestation alone and daily tides B. Basin rain, steep tributaries, sediment/channel mobility, drainage effects and floodplain exposure C. Glacier melt alone and absence of all embankments D. Urban paving in Guwahati alone

Answer: B.

Explanation: The correct model is coupled basin-channel-exposure risk. Single-factor answers fail.

Q7. Original - landslide threshold

Why can the same 24-hour rainfall total produce different landslide outcomes?

A. Rain totals never affect pore pressure. B. All slopes have identical geology. C. Intensity pattern, antecedent wetness, drainage, slope material and cutting differ. D. Earthquakes prevent rainfall-triggered slides.

Answer: C.

Explanation: Thresholds depend on both rainfall history and site susceptibility.

Q8. UPSC Prelims 2022, GS-I, Q62 - tea-producing states; key unavailable locally

Consider the following States:

1. Andhra Pradesh
2. Kerala
3. Himachal Pradesh
4. Tripura

How many of the above are generally known as tea-producing States?

A. Only one State B. Only two States C. Only three States D. All four States

INFERRED ANSWER - NOT OFFICIALLY VERIFIED: D. Confidence: high.

Elimination: Kerala and Himachal Pradesh are established tea-producing states; Tripura has a recognised tea sector; Andhra Pradesh has tea cultivation, including the Eastern Ghats/Araku context. All four therefore route to D. The official key was not available locally, so this package does not upgrade the inferred routing to official status.

Q9. Original - jhum

Which statement is most balanced?

A. Jhum outcomes depend on fallow length, customary tenure, crop diversity, slope and fire management. B. Every jhum field is sustainable regardless of shortened fallow. C. All jhum must be replaced by a single plantation crop. D. Burning has no nutrient or erosion effect.

Answer: A.

Explanation: Context matters. Long-fallow diversified systems and short-fallow degraded systems cannot be treated identically.

Q10. Original - tea geography

Why is rainfall alone insufficient to explain Assam tea?

A. Tea grows only in dry alkaline soil. B. Drainage, acidity, planting material, labour, processing speed, pests and markets also shape output and quality. C. Tea leaves can be stored unprocessed for months without quality loss. D. Altitude and temperature never matter.

Answer: B.

Explanation: Tea is a coupled agroecological and institutional system.

Q11. Original - urban flood

Which chain best explains Guwahati's pluvial/flash-flood problem?

A. Snowmelt -> sea-level rise -> tsunami B. Winter western disturbance -> desert dune migration C. Intense rain + hill runoff + sealed surfaces + wetland/drainage loss + constrained outfalls D. Brahmaputra bank erosion alone in every ward

Answer: C.

Explanation: Urban flooding has a local catchment and drainage pathway even when river level creates backwater.

Q12. Original - climate evidence

Which statement follows correct climate-evidence discipline?

A. One severe flood proves regional climate change. B. A national warming trend gives the exact rainfall trend at every Assam station. C. Any model projection is an observed fact. D. Event attribution requires a specific method; a trend or projection cannot by itself attribute one event.

Answer: D.

Explanation: Observation, trend, attribution and projection are distinct evidence classes.

Q13. Remedial - temperature

Why can frost occur within the wider humid-subtropical Northeast?

A. Elevation, cold-air drainage and individual weather events create local cold pockets. B. Humidity guarantees all temperatures stay above freezing. C. Köppen classes prohibit frost by definition. D. Frost requires a desert climate.

Answer: A.

Q14. Remedial - soils

Which statement is correct?

A. High rainfall makes every soil permanently sterile. B. Leaching may acidify hill soils, while floodplains can receive fresh alluvium and waterlogging creates other constraints. C. Forest biomass contains no nutrients. D. Red-yellow soils occur only in arid Rajasthan.

Answer: B.

Q15. Remedial - flood and waterlogging

Waterlogging differs from river inundation mainly because:

A. it always causes permanent bank loss; B. it occurs only on mountain summits; C. it reflects poor or blocked drainage and can persist after rainfall or river peak; D. it has no urban dimension.

Answer: C.

Q16. Remedial - infrastructure

Which is the best climate-resilient hill-road principle?

A. Maximise cut height to reduce length. B. Dump spoil into the nearest stream. C. Treat drainage as a maintenance issue only after opening. D. Align with susceptibility, design cross-drainage and slope support, manage spoil, monitor and maintain.

Answer: D.

Q17. UPSC Prelims 2024, GS-I, Set A, Q70 - officially verified

The North Eastern Council (NEC) was established by the North Eastern Council Act, 1971. Subsequent to the amendment of the NEC Act in 2002, the Council comprises which of the following members?

1. Governor of the Constituent State
2. Chief Minister of the Constituent State
3. Three Members to be nominated by the President of India
4. The Home Minister of India

Select the correct answer using the code given below:

A. 1, 2 and 3 only B. 1, 3 and 4 only C. 2 and 4 only D. 1, 2, 3 and 4

Answer: A - officially verified.

Exam takeaway: NEC is a statutory regional planning and coordination institution. For live scheme details, use the current NEC administrative source; do not infer resilience merely from project count.

Q18. Original - transboundary forecasting

What is the strongest benefit of upstream-downstream hydrometeorological data exchange?

A. It eliminates all forecast error. B. It can increase lead time and improve coordinated action, while local rain and failures remain uncertain. C. It stops channel migration. D. It converts every floodplain into dry land.

Answer: B.

Q19. Original - vegetation

Why does climate not determine one uniform forest type across the Northeast?

A. Plants do not respond to water. B. Latitude is identical everywhere. C. Altitude, soil, drainage, disturbance, succession and connectivity modify the climatic envelope. D. Bamboo occurs only above the tree line.

Answer: C.

Q20. Original - 2026 dashboard

Which is the valid interpretation of the NESAC/NERDRR report dated 10 August 2026?

A. It is the final Assam monsoon report. B. It proves exactly 99 villages were affected in the whole state for the entire season. C. It is a forecast issued before the rain. D. It is a satellite-coverage-bounded inundation status for the acquisition date.

Answer: D.

Rotation audit

Q1-Q20 keys: A, B, C, D, A, B, C, D, A, B, C, D, A, B, C, D, A, B, C, D.

B. Solved locally verified Mains PYQs

PYQ 1. UPSC GS-I 2021

Question: Differentiate the causes of landslides in the Himalayan region and Western Ghats.

Model solution

Claim: Landslides in both regions require susceptible slopes and a trigger, but the Himalaya is dominated by young tectonics, extreme relief and active river incision, whereas the Western Ghats more often combine deeply weathered plateau margins, escarpments and intense rain.

Named evidence/example: In the eastern Himalaya and Northeast hill corridors, fractured rocks, steep relief, earthquakes, high monsoon rainfall, toe erosion and aggressive road cutting interact. The Patkai-Naga-Mizo folds and Arunachal corridors also contain weak, weathered and structurally complex material. In the Western Ghats, lateritic/weathered mantles, steep escarpments, quarrying and intense orographic rainfall are prominent.

Analysis: Himalayan river incision and ongoing uplift continually rejuvenate relief. Seismic shaking creates fresh failures or cracks that later monsoon rain reactivates. Road cuts intercept subsurface flow, and blocked drains raise pore pressure. In the Western Ghats, saturation of thick regolith and failures along cut or quarry faces are frequent mechanisms.

Qualification: Neither belt has a single cause. Local lithology, antecedent wetness, land use and design quality may outweigh the regional generalisation at a particular site.

Verdict: Policy must shift from post-slide clearance to corridor-scale susceptibility mapping, rainfall thresholds, drainage, slope reinforcement/bioengineering, spoil control and monitored closures.

Why this earns marks: It differentiates the two regions causally, names Northeast processes, retains site-level qualification and closes with targeted risk reduction.

PYQ 2. UPSC GS-I 2021

Question: What are the environmental implications of the reclamation of water bodies into urban land use? Explain with examples.

Model solution

Claim: Reclamation converts flexible ecological storage into rigid exposure, increasing urban flood peaks while removing water-quality and habitat functions.

Named evidence/example: In Guwahati, the interaction of hills, wetlands, channels and the Brahmaputra makes natural storage and drainage continuity crucial. Encroachment or fragmentation of beels, sealed surfaces, hill

sediment and waste-blocked drains increase ponding and flash-flood risk.

Analysis: Water bodies attenuate runoff, receive sediment, recharge groundwater in suitable settings, support fisheries and moderate microclimate. Filling them displaces water rather than eliminating it. The resulting flow enters roads and houses faster; constrained outfalls during high river stage prolong waterlogging. Habitat connectivity and livelihood access decline.

Qualification: Not every stagnant or polluted urban depression performs all functions equally, and restoration must address sewage and invasive species. Planned development may require limited intervention, but hydrological compensation must be demonstrated at catchment scale.

Verdict: Cities need mapped flow paths, protected wetland networks, drain-wetland connectivity, pollution control, permeable surfaces and enforceable development setbacks.

Why this earns marks: It gives environmental mechanisms rather than a generic list, uses Guwahati appropriately and offers a qualified planning solution.

PYQ 3. UPSC GS-I 2020

Question: Account for the huge flooding of million cities in India including the smart ones like Hyderabad and Pune. Suggest lasting remedial measures.

Model solution

Claim: Large-city flooding is a drainage-basin failure created when extreme rainfall meets sealed surfaces, lost storage, obstructed drains and vulnerable land use.

Named evidence/example: Guwahati shows the same logic in a humid subtropical setting: short intense rain and hill runoff converge on lowlands; wetland loss, drain encroachment, sediment/waste and high receiving-water level can prolong flooding.

Analysis: Urbanisation shortens lag time and raises peak discharge. Roads and walls redirect flows, undersized culverts create bottlenecks, and building in low corridors converts a weather event into asset loss. Smart sensors cannot compensate for a physically broken drainage network.

Qualification: Design standards must reflect local rainfall data and climate uncertainty; river flooding, pluvial flooding, flash runoff and sewer surcharge should not be merged into one category.

Verdict: Use catchment-scale drainage plans, protect blue-green storage, maintain drains before monsoon, enforce flood-compatible zoning, retrofit culverts, manage solid waste and connect impact forecasts to ward action.

Why this earns marks: It explains the urban hydrograph, transfers the mechanism to Guwahati, distinguishes flood types and proposes structural plus governance reforms.

PYQ 4. UPSC GS-I 2023

Question: Discuss the consequences of climate change on the food security in tropical countries.

Model solution

Claim: Climate change threatens food security through availability, access, utilisation and stability, with humid tropical/subtropical regions facing compound heat, flood, pest and infrastructure risks.

Named evidence/example: Northeast rice systems face submergence, erosion, sand casting and delayed field operations; hill horticulture faces road closures and post-harvest loss; tea faces heat-moisture stress and pest/quality risks. IMD agromet advice dated 18 August 2026 warned that heavy rain could submerge crops, cause root rot and wash nutrients, while advising field drainage.

Analysis: Extreme rain can destroy standing crops while longer dry gaps reduce establishment. Heat affects labour and livestock. Transport interruption raises food prices and reduces market access. Diet quality falls when diversified local systems are replaced by a narrow cash crop without storage or market security.

Qualification: Climate is not the sole determinant; seed systems, irrigation/drainage, tenure, credit, roads, storage and public distribution mediate outcomes. A warning is not an observed crop-loss estimate.

Verdict: Adaptation requires resilient varieties, seed reserves, soil-water management, diversified farming, agromet services, storage, reliable connectivity and social protection.

Why this earns marks: It uses the four pillars of food security, a bounded current example, regional specificity and a no-single-cause qualification.

PYQ 5. UPSC GS-I 2023

Question: Identify and discuss the factors responsible for diversity of natural vegetation in India. Assess the significance of wildlife sanctuaries in rainforest regions of India.

Model solution

Claim: India's vegetation diversity arises from the interaction of temperature, moisture, altitude, soil, drainage and disturbance, while rainforest sanctuaries protect ecological processes that exceed individual species.

Named evidence/example: Northeast India compresses tropical evergreen, semi-evergreen, moist deciduous, subtropical broadleaf/pine and montane transitions across short horizontal distances. Floodplain grasslands, beels, bamboo-cane systems and Eastern Himalayan forests add habitat diversity.

Analysis: Altitude lowers temperature; slope orientation changes moisture; flood and landslide create succession; soil and drainage filter species; jhum, fire, roads and plantations reshape the mosaic. Sanctuaries secure habitat, watersheds, pollination, carbon, genetic diversity and climate refugia.

Qualification: Isolated protected "islands" cannot sustain all ecological movement. Conservation must include corridors, community rights, invasive-species management and compatible livelihoods outside boundaries.

Verdict: Rainforest protection should be landscape-based: core habitats plus river, forest and elevation connectivity, participatory buffers and monitored development corridors.

Why this earns marks: It links vegetation to processes, uses a regionally rich example and moves beyond sanctuary notification to connectivity and rights.

PYQ 6. UPSC GS-I 2023

Question: Discuss the meaning of the statement that the wind known as Purvaiya in the Bhojpur region has a strong influence on the cultural ethos of the region.

Model solution

Claim: Seasonal winds become cultural phenomena when their arrival reorganises agricultural time, sensory experience, labour, mobility and collective memory.

Named evidence/example: The Purvaiya example can be compared with Assam's Bordoisila: a pre-monsoon storm tradition associated with squalls, lightning, relief from heat, crop effects and local narratives.

Analysis: Farmers read wind, cloud and humidity as a practical calendar. Songs and sayings encode anticipation, fear and seasonal change. Architecture, work timing and festivals respond to recurring weather. Thus climate is not an external background; it enters language and identity.

Qualification: Cultural meanings vary by community and generation, and scientific names should not erase local ecological knowledge. Conversely, folklore is not a substitute for forecast verification.

Verdict: Regional geography is strongest when it connects circulation and seasonality to lived landscapes without romanticising risk.

Why this earns marks: It directly interprets "cultural ethos," adds a relevant Northeast comparison and balances local knowledge with scientific evidence.

C. Original solved Mains practice

Practice 1 - 10 marks, 150 words

Question: Explain why Northeast India should be described as a humid-subtropical mosaic rather than a single climatic region.

Model solution

Claim: “Humid subtropical” captures the hot-wet summer and mild/cool winter tendency of many lowland sectors, but relief divides Northeast India into sharply contrasting climatic units.

Named evidence/example: The Brahmaputra and Barak valleys, southern Meghalaya escarpment, plateau interior, Patkai-Naga-Mizo folds and Arunachal high mountains differ in elevation, slope orientation and exposure to Bay moisture.

Analysis: Monsoon inflow is uplifted on windward slopes, channelled through gaps and partly reduced leeward. Valley continentality, winter inversion and fog contrast with cooler plateaus and montane frost. Tropical monsoon, Cwa/Cfa-like and highland conditions can therefore occur within the wider region.

Qualification: Köppen codes apply to station thresholds, not administrative regions; “China type” is an analogy, not a complete classification.

Verdict: The correct regional unit is a circulation-linked but terrain-differentiated climatic mosaic.

Why this earns marks: It defines, maps, explains and qualifies the label within the word limit.

Practice 2 - 15 marks, 250 words

Question: “Brahmaputra floods are simultaneously geomorphic processes, ecological events and social disasters.” Analyse.

Model solution

Claim: Brahmaputra flooding cannot be reduced to excess rainfall because water, sediment, channel change, ecology and exposure operate together.

Named evidence/example: Steep Himalayan and hill tributaries deliver rapid runoff and sediment to a wide braided/anabranching Assam reach with chars, beels and mobile banks. NESAC/NERDRR’s 10 August 2026 satellite report mapped 2,155.17 ha inundation in covered portions of Darang, Goalpara and Kamrup Rural, mainly agricultural land.

Analysis: Geomorphically, flood pulses build and erode bars, shift channels and attack banks. Ecologically, seasonal inundation connects wetlands, supports fisheries and renews soil moisture/sediment. Socially, the same processes destroy crops, homes, roads and tenure security when settlements occupy active corridors or when embankments and drainage fail. Erosion differs from inundation because it may permanently remove land; waterlogging differs because drainage remains blocked after peaks.

Qualification: The satellite figure is partial and date-specific, not a statewide seasonal total. Deforestation may alter individual catchments but cannot alone explain basin-scale flood behaviour. Snow/cryosphere contribution is event- and season-dependent.

Verdict: Policy must combine forecasting, wetland storage, selective embankments, flood-compatible land use, erosion compensation/resettlement and transboundary data rather than promise complete flood elimination.

Why this earns marks: It integrates physical and human geography, uses bounded current evidence, distinguishes three processes and gives a portfolio response.

Practice 3 - 20 marks, 300 words

Question: Evaluate whether road-led connectivity can be made compatible with slope stability, drainage and livelihood resilience in Northeast India.

Model solution

Claim: Connectivity is indispensable for health, markets and strategic integration, but in high-rainfall mountains its benefits depend on whether road design treats slope and drainage as a living system.

Named evidence/example: NEC's FY 2025-26 release recorded Rs 1,496.78 crore expenditure, including Rs 704.66 crore under NESIDS-Roads, while NHIDCL's June 2026 NH-717A procurement explicitly included slope protection, landslide mitigation, reconstruction and rockfall barriers.

Analysis: Roads can lower travel time, reduce post-harvest loss and improve disaster access. Yet over-steep cuts remove toe support, intercepted subsurface flow raises pore pressure, undersized culverts concentrate runoff, and spoil dumping chokes streams. Failures then isolate settlements precisely during emergencies. New roads may also fragment forest and induce construction in hazard corridors.

Compatibility requires:

1. corridor-scale geological and hydrological investigation before alignment;
2. avoidance of old slides and critical habitat where alternatives exist;
3. staged excavation, benches, retaining and bioengineering matched to material;
4. continuous lined/unlined drainage with safe outlets and cross-drainage;
5. spoil sites outside channels, monitored for stability;
6. rainfall thresholds, sensors and pre-emptive closure;
7. maintenance contracts, local reporting and redundant links;
8. livelihood safeguards, fair land processes and benefit monitoring.

Qualification: Project sanction, expenditure or completion does not prove resilience. The NH-717A example is from the eastern Himalayan edge and does not establish performance across Assam or all Northeast geology.

Verdict: Road-led connectivity is compatible with resilience only when lifecycle drainage, slope monitoring, ecological safeguards and local access outcomes are treated as core project outputs rather than mitigation appendices.

Why this earns marks: It acknowledges development benefits, traces the full failure mechanism, names current official evidence, states its limit and provides implementable criteria.

Practice 4 - 15 marks, 250 words

Question: Examine jhum as a socio-ecological system and suggest context-sensitive transition pathways.

Model solution

Claim: Jhum is a rotational land-use institution combining mixed cropping, fire, fallow succession and customary tenure; its sustainability depends more on cycle conditions than on the label itself.

Named evidence/example: Long enough fallow restores biomass, litter and nutrient cycling, while diversified crops distribute risk. ICAR's July 2026 case converted 0.2 ha of old jhum land in Arunachal Pradesh to a managed Assam lemon orchard with intercropping.

Analysis: Shortened fallow can reduce recovery, increase weeds and erosion, and create repeated fire pressure. However, compulsory replacement may undermine food diversity, common rights and cultural identity, while exposing households to a single volatile market. Transition options include improved fallow, assisted natural regeneration, community fire lines, agroforestry, horticulture, terrace/water management, local processing and secure market access.

Qualification: The ICAR case is a bounded demonstration, not a universal prescription. Suitability depends on slope, water, tenure, labour, diet, market distance and community consent.

Verdict: Policy should move from "eradication of jhum" to negotiated landscape planning that measures soil cover, fallow recovery, income stability, food diversity and rights.

Why this earns marks: It avoids condemnation and romanticisation, explains the cycle and gives consent-based, measurable alternatives.

Practice 5 - 10 marks, 150 words

Question: Distinguish FACT, STATUS, FORECAST and INFERENCE using current Northeast hydroclimate evidence.

Model solution

Claim: Evidence labels prevent a current-affairs example from becoming an exaggerated geographical conclusion.

Named evidence/example: FACT: MoES reports India-scale warming of about 0.7 degrees C during 1901-2018. STATUS: NESAC/NERDRR mapped inundation in covered Assam areas on 10 August 2026. FORECAST: IMD's bulletin issued 18 August warned of isolated heavy rain and thunderstorm/lightning for 19 August. INFERENCE: further intense rain over wet catchments could increase flood and slope risk.

Analysis: Facts describe stable observations; status reports describe realised conditions at a date and coverage; forecasts are probabilistic and validity-bounded; inferences connect evidence through a stated mechanism.

Qualification: The flood map was partial, the IMD warning did not itself prove realised impact, and neither established a final 2026 monsoon outcome.

Verdict: UPSC answers should state source, date, spatial coverage and evidence class before drawing policy lessons.

Why this earns marks: It uses four labels accurately, attaches current examples and explicitly prevents forecast-to-outcome and event-to-climate errors.

D. Answer-writing drills

Map drill

In 90 seconds draw:

1. a broad Brahmaputra Valley west-east;
2. Barak Valley to the south;
3. Meghalaya Plateau and its southern escarpment;
4. Patkai-Naga-Manipur-Mizo hills;
5. Arunachal foothills/high mountains;
6. Bay-moisture arrows through Bangladesh;
7. windward uplift and one leeward pocket;
8. tributary arrows to Brahmaputra and Barak.

Ten named evidence units

1. Khasi-Jaintia southern escarpment - orographic uplift.
2. Brahmaputra chars and beels - channel and floodplain dynamism.
3. Barak-Surma-Kushiyara - transboundary lowland drainage.
4. Bordoisila - pre-monsoon convection and cultural geography.
5. Guwahati wetlands and hill runoff - urban compound flooding.
6. IMD Bulletin 66/2026 - validity-bounded forecast.
7. NESAC/NERDRR 10 August 2026 - partial satellite status.
8. Tea Board Jan-Apr 2026 - provisional sector evidence.
9. NEC/NHIDCL 2026 - implementation/design status with outcome caveat.
10. ICAR old-jhum lemon case - bounded transition example.

Common examiner deductions

- One Köppen code for the whole Northeast.
 - “Mawsynram is wet because mountains block clouds” without circulation and orientation.
 - Flood, erosion and waterlogging used interchangeably.
 - Deforestation presented as the sole flood cause.
 - Alluvial and hill soils treated as uniformly infertile.
 - Jhum described as primitive without fallow or tenure.
 - Project spending presented as resilience success.
 - A forecast presented as realised damage.
 - One flood season presented as climate attribution.
 - A Mains conclusion that repeats the introduction without a qualified verdict.
-